Module 1
Foundations of Multimodal AI

Module 2
Integrating Visual and Video Modalities

Image Generation and Captioning

- Understanding Image Captioning with Meta's Llama
  Video • 7 min
- Reading: Introduction to Text-to-Video and Image-to-Video Technologies
  Ungraded Plugin • 12 min
- Demo: Text-to-Video Generation with OpenAI's Sora
  Video • 7 min
- Lab: DALL-E Image Generation Guide for Beginners
  Ungraded App Item • 20 min
- Reading: Strengths, Limitations, and Practical Applications of Multimodal Vision Models in Real World Scenarios
  Ungraded Plugin • 8 min
- Lab: Build an Image Captioning System with watsonx and IBM's Granite
  Ungraded App Item • 30 min
- Image Generation and Captioning
  Practice Assignment • Grade: 100%

Module Summary and Evaluation

- **Reading: Summary and Highlights**
  Reading • 3 min
- Cheat Sheet: Integrating Visual and Video Modalities
  Ungraded Plugin • 15 min
- Graded Quiz: Integrating Visual and Video Modalities
  Graded Assignment • 21 min

Module 3
Advanced Multimodal Applications

Reading: Summary and Highlights

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Congratulations! You have completed this module. At this point, you know that:

Image captioning with Meta's Llama

- Combining computer vision with natural language processing creates powerful tools for understanding visual content.
- Three main stages of the image captioning process with a multimodal large language model (LLM) are:
  - Input processing
  - Image validation and encoding
  - Multimodal LLM processing
- Input processing receives and prepares the image and optional text prompt.
- Image validation and encoding validate and convert the image into a format (e.g., Base64) suitable for the model.
- Multimodal LLM processing combines visual and textual information to generate a descriptive caption.
- Core components of the image captioning system to produce captions tailored to prompts are:
  - Visual encoders
  - Text embedding
  - Fusion layers
  - Language generation tools
- Implementing an image captioning system using Meta's Llama 4 Maverick model via IBM watsonx involves:
  - Importing libraries and authenticating access
  - Encoding images and preparing prompts
  - Sending combined image-text messages to the model
  - Extracting descriptive text from the model's response

Text-to-video generation with OpenAI's Sora

- Sora is a multimodal, diffusion-based transformer model developed by OpenAI that can generate high-quality video from text or image inputs.
- For accurate results, you must craft a structured prompt and include essential elements, such as scene context, visual details, and motion required in your clip.
- Steps for text-to-video generation:
  - Open your browser and go to sora.openai.com to access the official Sora interface.
  - If not logged in, click "Log In" or "Sign Up" for a new OpenAI account.
  - After logging in, you'll land on the "Explore page," where you can browse others' videos for inspiration.
  - Use the composer at the bottom to enter your text prompt describing the video you want.
  - Before creating, review your settings:
    - Choose Type: Video
    - Set aspect ratio, resolution, duration, number of variations, and style preset
    - Options depend on your OpenAI subscription tier
  - Click "Create video" to submit your request; processing takes 30 seconds to a few minutes
    - Finished videos appear under "My Videos"
    - Hover to "Preview"
    - Click to open in a lightbox and use the arrow keys to view variations
- Select a variation to refine using the editing toolbar:
  - "Edit" storyboard or recut clips
  - Use "Remix" to describe changes in natural language
  - Use "Blend" to merge with another video
  - Use "Loop" to create seamless repeats
- After editing, a new variation is added to your set.

Mark as completed